

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 430 — Past Questions & Materials

by Renance

Course: EMT 430

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 400

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Paper 1 of 2 — EMT 430 ENVIRONMENT MANAGEMENT ^ TOXICOLOGY

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT & TOXICOLOGY
EMT 430 (ENVIRONMENTAL AND PETROLEUM MICROBIOLOGY)
2019/2020 ACADEMIC SESSION 1ST SEMESTER EXAMS
ANSWER TWO QUESTIONS FROM EACH SECTION

USE SEPARATE BOOKLETS FOR THE TWO SECTIONS

CREDIT: 2 UNITS

TIME ALLOWED: (2 HOURS)

SECTION A

1. Explain the factors that influence the microbial diversity and populations
2. Write concisely on: i) Ecosystems ii) Extremophiles iii) "Spaghetti bacterium"
iv) *Theomargaritana mebiensis*
3. Write briefly on the following: a) Self-purification of water b) Describe the process of purification of water. c) State three (3) causes and effects of water pollution.
d) Waterborne diseases. Give an example each caused by bacteria, virus, protozoan and worms.

SECTION B

- 4a) Explain briefly five [5] factors influencing bioremediation of petroleum polluted sites.
- 4b) Highlight two [2] methods of preventing microbial corrosion of oil field equipment.
- 4c) Enumerate five [5] consequences of microbial deterioration of petroleum and its products.
- 5a. Identify the major sources of airborne microorganisms
- 5b. List the various types of airborne diseases, their symptoms and prevention and control with emphasis on Measles, Whooping cough, *Diphtheria*
- 5c. In a tabular form differentiate between pathogen transmitted via droplet and pathogen transmitted via airborne
- 6a. As an experienced hire in an oil facility explain to the foremen the possible causes of (SOURING) stating at least five problems that may arise as a result of its occurrence in their facility
- 6b State and explain at least four classes of microorganisms with examples that maybe responsible for deterioration of the oil facility in their location
- 6c. Advise generally, with examples ways to curb the menace of souring in their facility

Proteobacteria
Cyanobacteria
Candida

Flavobacterium

Sulfur *Acid*
of

membrane
membrane
membrane

typical
fuel
add

Paper 2 of 2 — EMT 430 Past Questions - Environmental and Petroleum Microbiology

Level 400 • First Semester • PastQuestion

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- 6b State and explain at least four classes of microorganisms with examples that maybe responsible for deterioration of the oil facility in their location

6c. Advise generally with examples ways to curb the menace of souring in their facility